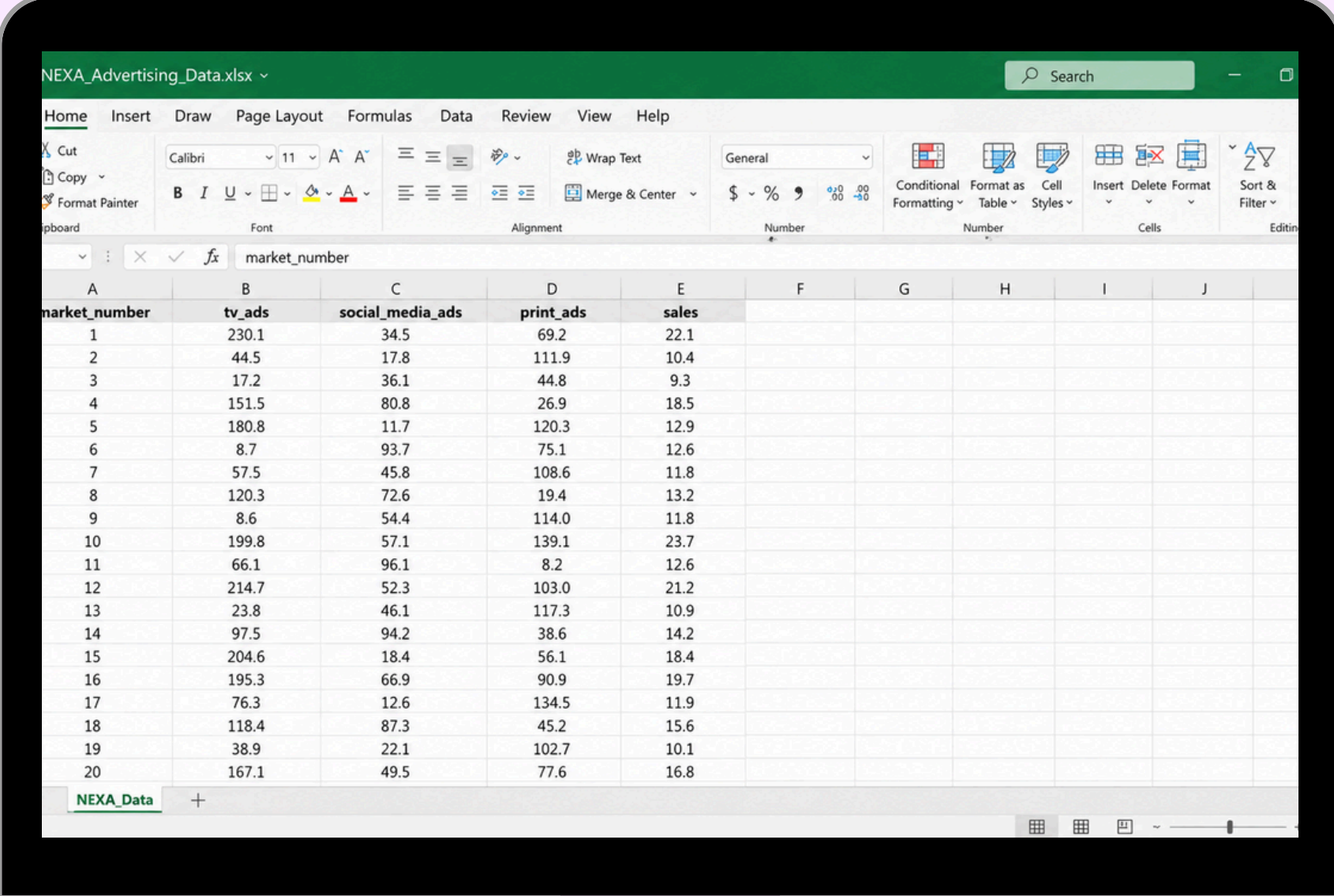


NEXA ONE: ↗

Rethinking the Advertising Mix

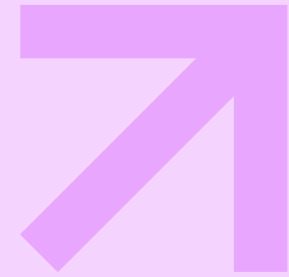
An analytics-led study of advertising spend, channel performance and sales across 200 markets.

By: Sumaiya Afreen
24f2006233



market_number	tv_ads	social_media_ads	print_ads	sales
1	230.1	34.5	69.2	22.1
2	44.5	17.8	111.9	10.4
3	17.2	36.1	44.8	9.3
4	151.5	80.8	26.9	18.5
5	180.8	11.7	120.3	12.9
6	8.7	93.7	75.1	12.6
7	57.5	45.8	108.6	11.8
8	120.3	72.6	19.4	13.2
9	8.6	54.4	114.0	11.8
10	199.8	57.1	139.1	23.7
11	66.1	96.1	8.2	12.6
12	214.7	52.3	103.0	21.2
13	23.8	46.1	117.3	10.9
14	97.5	94.2	38.6	14.2
15	204.6	18.4	56.1	18.4
16	195.3	66.9	90.9	19.7
17	76.3	12.6	134.5	11.9
18	118.4	87.3	45.2	15.6
19	38.9	22.1	102.7	10.1
20	167.1	49.5	77.6	16.8

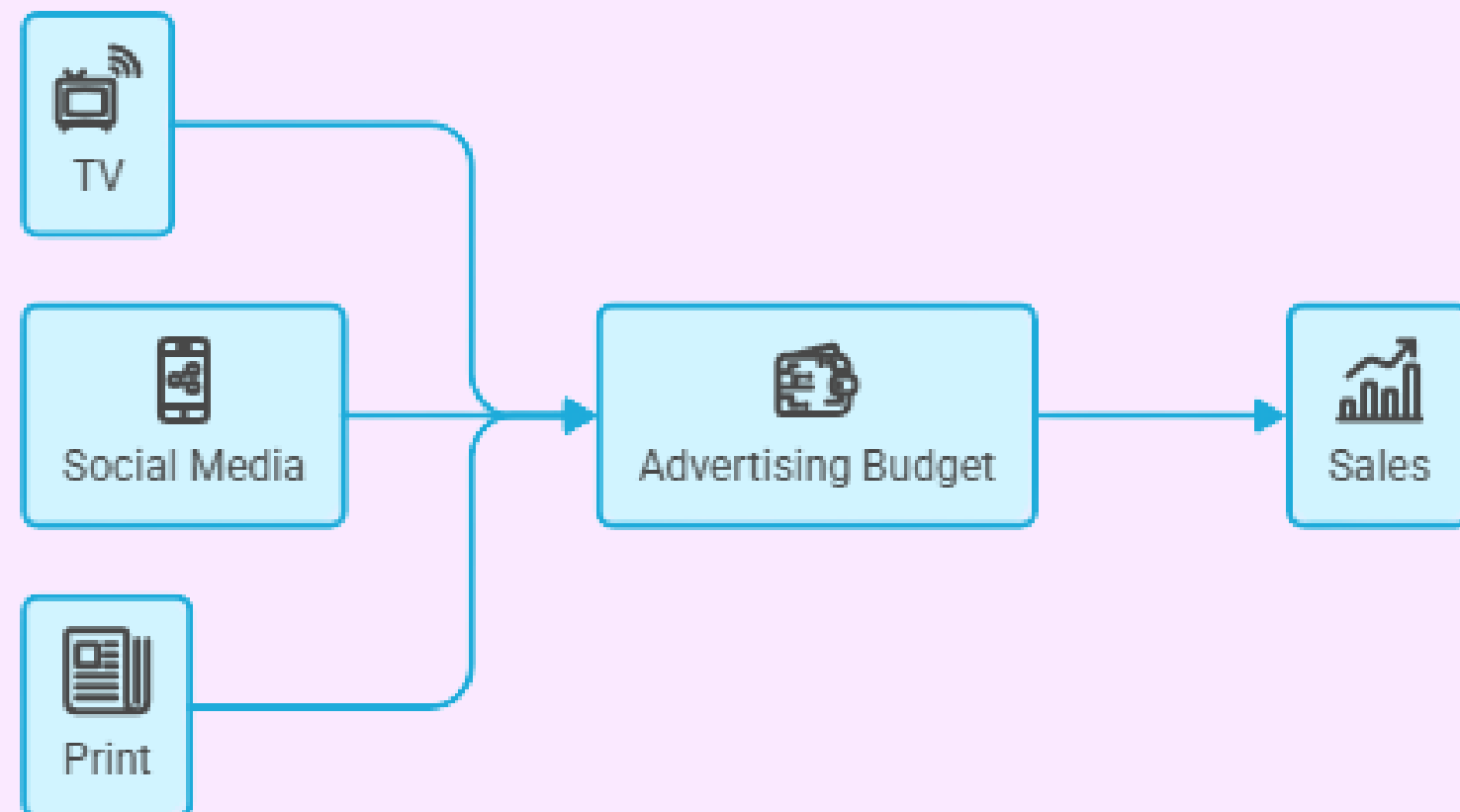
Business Problem



The Business Challenge

NEXA invests across TV, Social Media and Print, but total advertising spend alone does not show which channels contribute most to observed sales.

NEXA's Advertising Allocation Challenge



My analysis focuses on five questions

- Does higher advertising spend relate to higher sales?
- Which channel has the strongest association with sales?
- What do consumers value when choosing wireless noise-cancelling headphones?
- How are market and technology trends affecting the category?
- How should NEXA allocate its next-period advertising budget?

Analytical Approach ↗

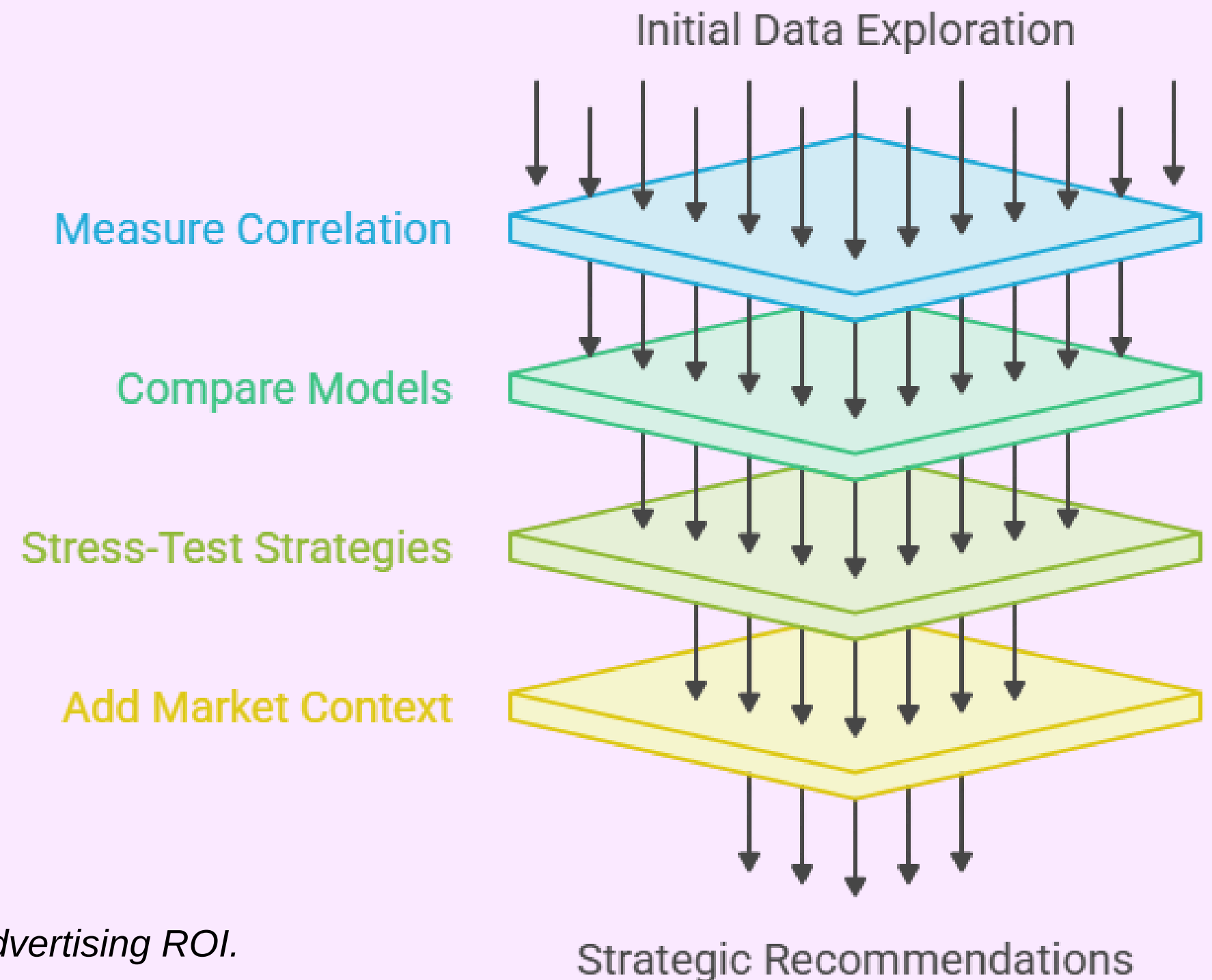
Methodology

The analysis combines statistical modelling with external market research to translate observed relationships into a practical advertising strategy.

- Pearson correlation → measures channel-sales association.
- OLS regression → evaluates total spend and individual channels.
- Standardized β → compares relative channel strength.
- Budget scenario → tests a potential Print → Social shift.
- Market research → adds consumer, technology and supply-chain context.

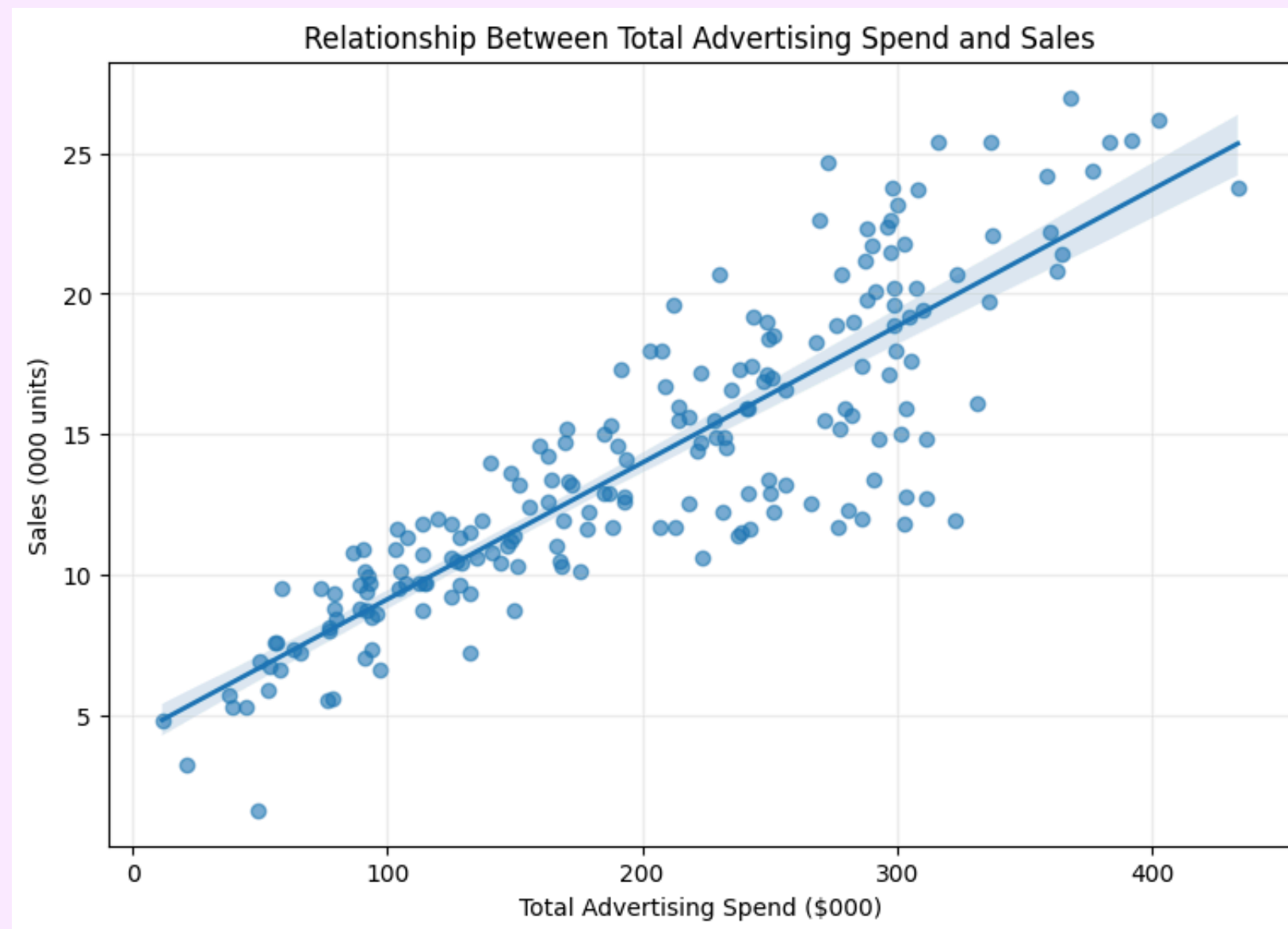
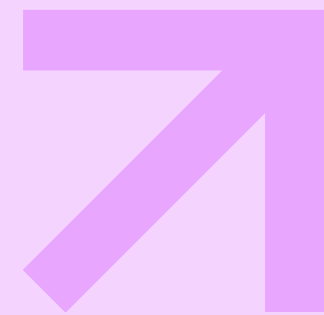
Note: Observational dataset: results indicate association, not causal advertising ROI.

Data-Driven Marketing Strategy Funnel



Key Finding 1

Advertising spend and sales move strongly together



Interpretation

The analysis shows a strong positive relationship between total advertising expenditure and sales.

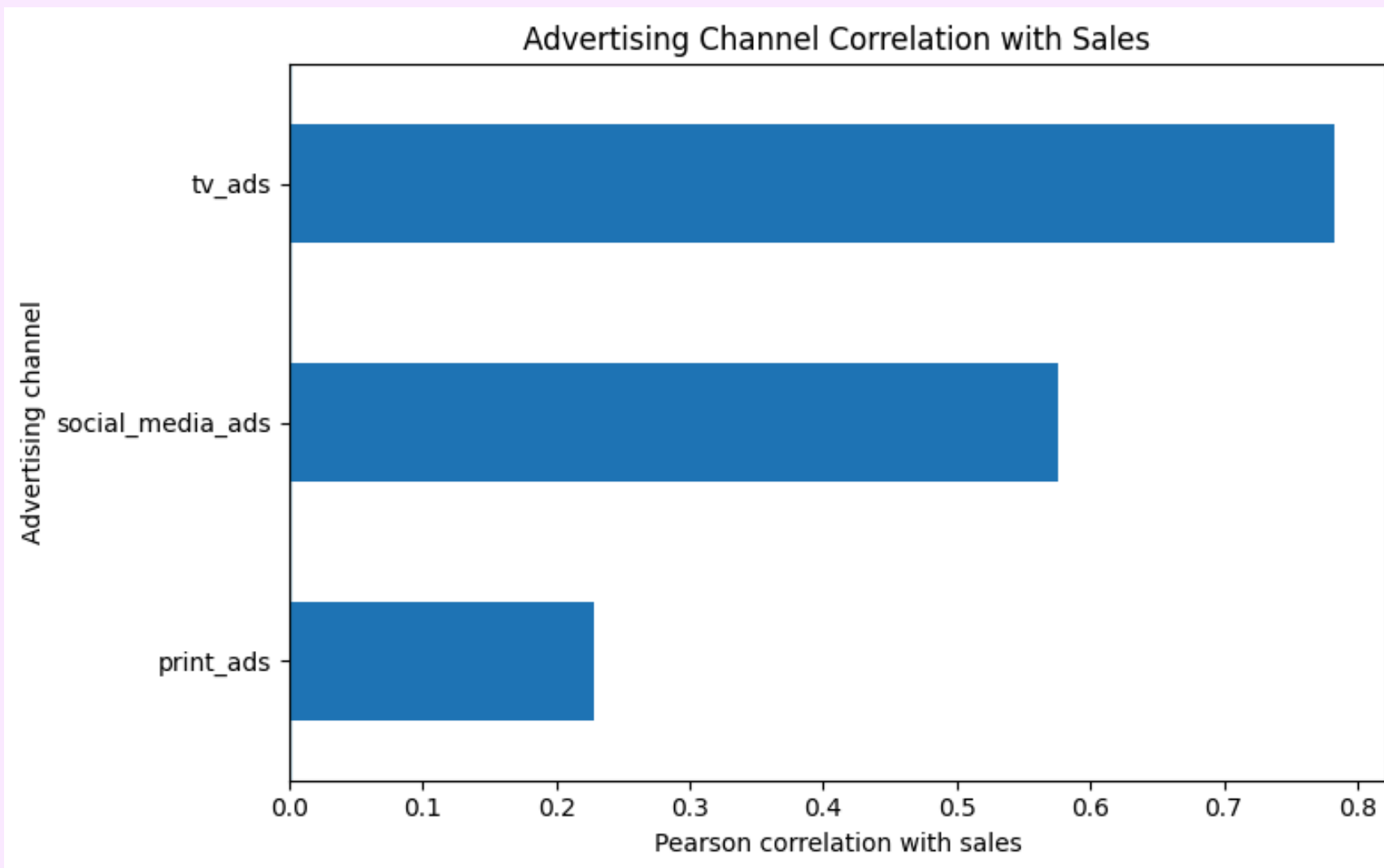
- Pearson correlation = 0.868, indicating a strong positive association.
- $R^2 = 75.3\%$, meaning total advertising expenditure explains 75.3% of the observed variation in sales.
- The advertising coefficient is 0.0487 and is statistically significant ($p < 0.001$).
- The estimated relationship is:

$$\text{Sales} = 4.243 + 0.0487 \times \text{Total Advertising Spend}$$

Business Insight: Markets with higher total advertising expenditure generally recorded higher observed sales..

Key Finding 2

The channel mix adds more information than total spend alone



Interpretation

The multichannel model provides more explanatory power than the total-spend model.

- R^2 increases from 75.3% to 89.7% when channels are modelled separately.
- TV has the strongest association ($r = 0.782$; $\beta = 0.753$).
- Social Media also shows a meaningful positive association ($r = 0.576$; $\beta = 0.537$).
- Print shows weak association and limited additional explanatory power ($r = 0.228$; $\beta = -0.004$; $p = 0.860$).

Business Insight: TV remains the strongest observed channel, while Social Media presents a potential opportunity and Print warrants reassessment.



Interpretation

The current advertising budget is heavily concentrated in TV, while Social Media has a smaller share despite its positive association with sales.

- TV: 73.21% of budget
- Social Media: 11.58%
- Print: 15.21% with limited explanatory power
- A \$10K Print → Social Media shift gives a model-implied increase from 14.022K to 15.918K sales units.

Recommended Action: Run a controlled market-level test before making a full-scale budget reallocation.

Strategic Opportunity



	Average Spend (\$000)	Budget Share (%)
tv_ads	147.04	73.21
social_media_ads	23.26	11.58
print_ads	30.55	15.21

Business Insight: The analysis suggests a potential opportunity to test a partial shift from Print toward Social Media rather than increasing the overall advertising budget.



Interpretation

- Consumers prioritize sound quality, battery life and comfort.
- High price sensitivity makes value for money important.
- The audio category is evolving, with newer formats gaining traction.
- Supply-chain pressures can affect costs and product availability.

Consumer Priorities are:

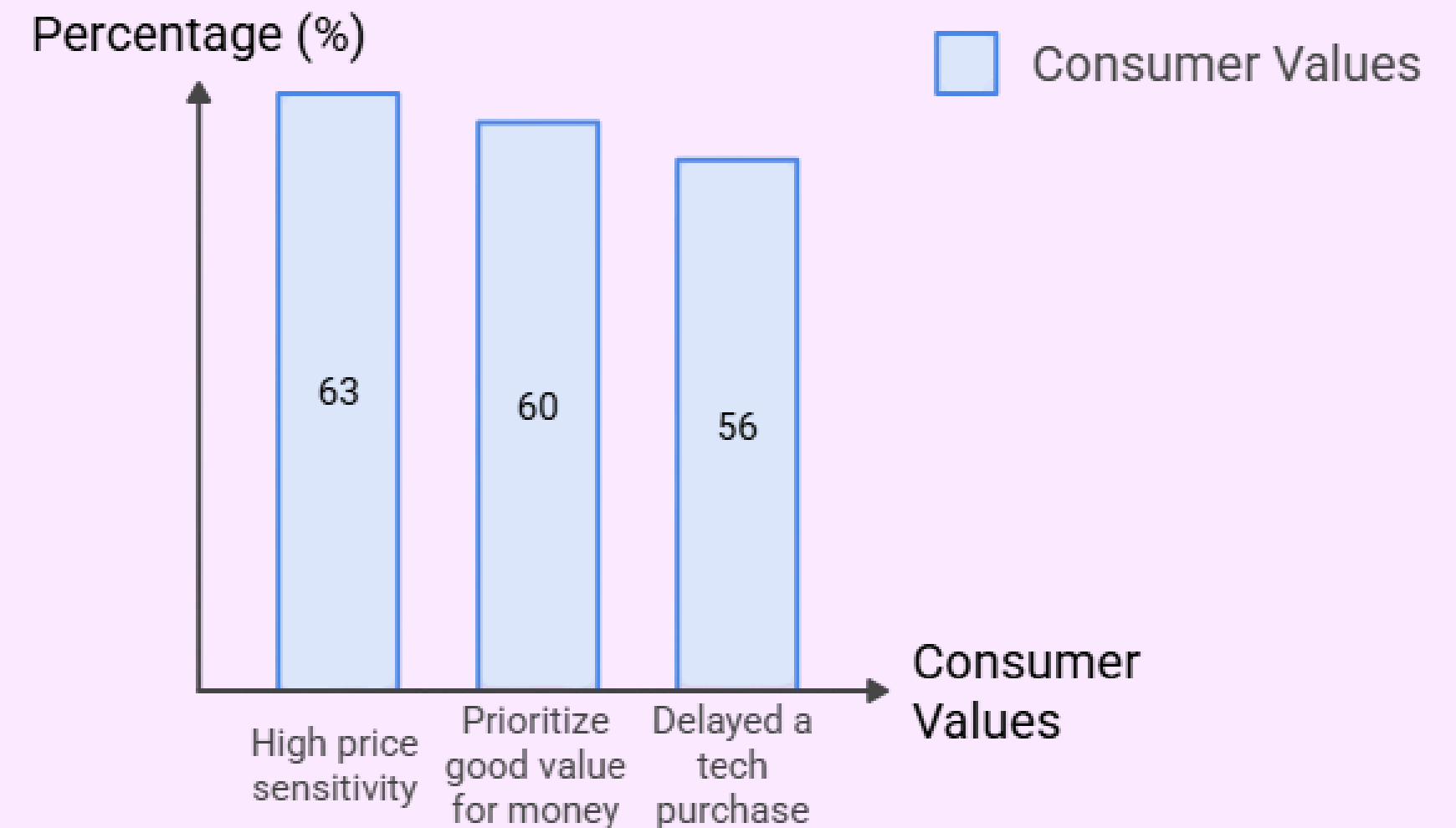


Business Insight:

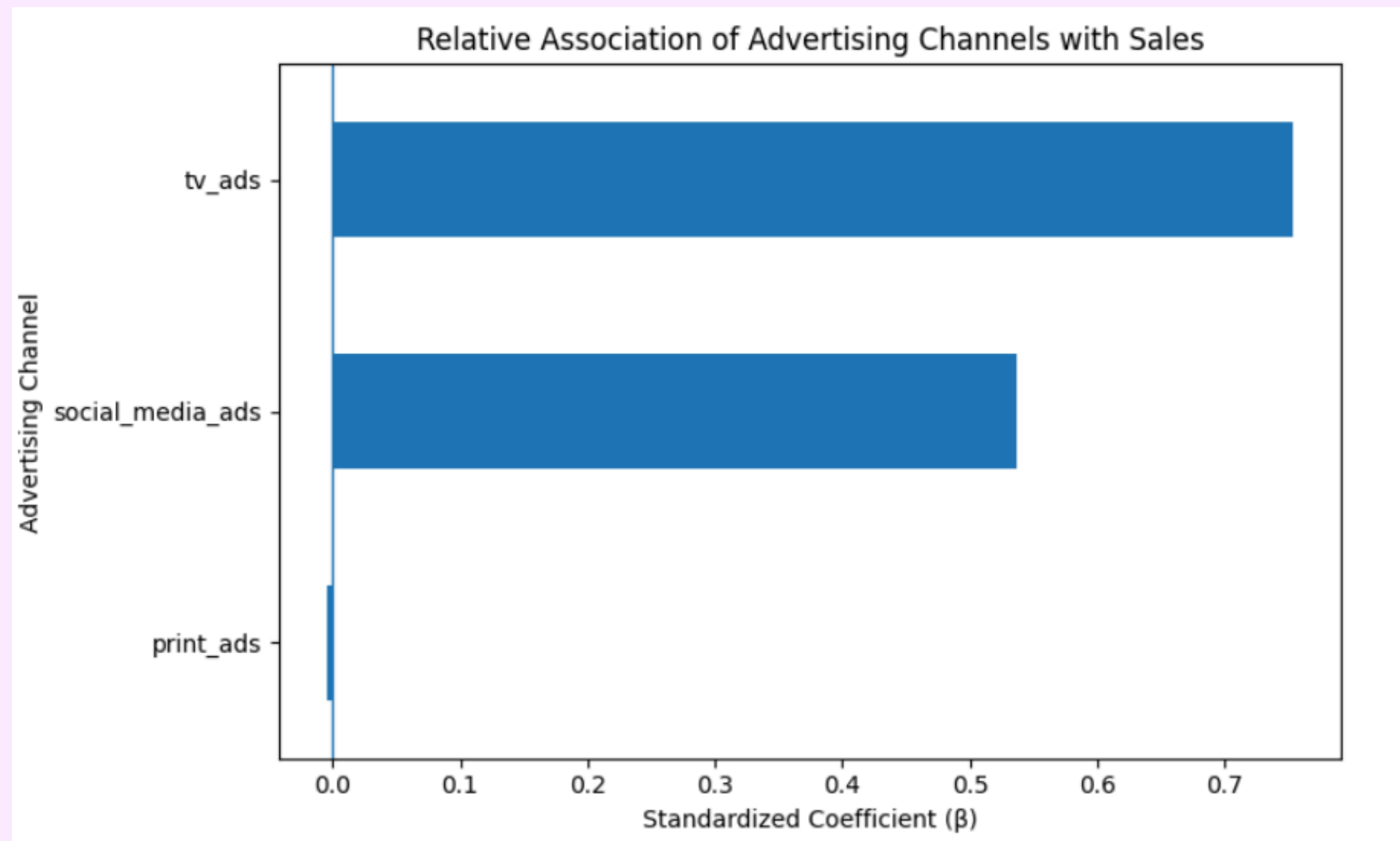
NEXA should choose the right channels, communicate clear product benefits, and adapt to changing consumer and market trends.

Market and Consumer Insights

Consumer Values Regarding Price



Recommendations ↗



Business Takeaway:

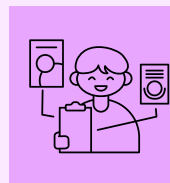
NEXA should not simply spend more. It should protect the channels that show strong evidence, test where the mix can improve, and communicate the product benefits consumers value most.



1. Maintain TV as a core channel

TV has the strongest observed relationship with sales ($r = 0.782$; $\beta = 0.753$).

Keep its role while continuing to monitor performance.



2. Test a Print → Social shift

Social Media shows a stronger observed association with sales than Print.

Test a partial reallocation before making a permanent change.



3. Make the product benefit clear

Build campaigns around:

Sound Quality | ANC | Battery | Comfort |
Durability | Value



4. Measure before scaling

Use controlled market tests to measure results before expanding the new mix.

